Commensurability of Observation Protocols: When Coherent Windows Determine a Process

Patrick S. Mahon*

Abstract

Observing a process through overlapping finite windows yields a family of local statistics. When are locally coherent window-statistics guaranteed to arise from a single global process? We call an observation protocol *commensurable* when empirical adequacy (pairwise coherence of the windows, EA) forces probabilistic realisability by a declared class (PR($\mathcal{R}$)); the failure — coherent-but-unrealisable data — is contextuality in the sense of Abramsky–Brandenburger, arrived at from the observational side. We recast the question geometrically: a protocol is commensurable iff its coherence polytope C equals its realisability polytope R, so contextuality is an exactly computable separation and every failure carries a rational witness. Within this frame we prove a structure theory for protocols whose contexts are windows on a symbolic dynamical system. The main results are: a *trichotomy* (acyclic reporting is commensurable for every language; full-support coordinate protocols are commensurable by the product measure; protocols carrying a constraint variety realise every rational polytope, hence admit no finite classification, by De Loera–Onn); a *parity theorem* (the golden-mean shift observed around a ring of length L is commensurable iff L is even, the odd-ring gap being a single extreme point, its own PR-box); a catalogue of ten *taming* mechanisms, each a structural obstruction to contextuality; and, assembled from these, *Circuit Localization* for the symmetric loopless marginal-determined languages — there commensurability is decided entirely by acyclicity and the tamings, contextuality excluded by structure rather than by search (the loopy and general-cyclic extensions are verified to finite scale and stated as such). The load-bearing combinatorial discriminant (a layered-ring walk is a simple cycle iff layer-injective) is machine-verified in Lean 4. We state the remaining open case — a universal impossibility conjecture for the strongly-connected non-symmetric class — together with its reduction to a combinatorial pruning lemma and the precise gap that reduction currently carries.

1 Introduction

A process unfolding in time is often accessible only through *windows*: one records the joint behaviour of finitely many coordinates over finitely many steps, then of others, in overlapping finite views. Each window yields a local probability table. The tables may agree wherever they overlap and yet be generated by no single underlying process — the observational face of quantum contextuality [2]. This paper asks, for symbolic dynamical observation, exactly when local coherence forces global realisability, and answers it for a large structured class.

We fix vocabulary (Paper II, now doing classification work). A finite alphabet A and finite supports $F \subseteq \mathbb{Z}$ give *contexts*, each a partition of A^F; a *protocol* is a family of contexts. Window-data is a consistent assignment of probabilities to the cells of each context.

Definition 1.1 (EA, PR($\mathcal{R}$), commensurability)**.** Window-data is *empirically adequate* (EA) if it is pairwise coherent: the contexts agree on the common subalgebra of each pair. It is *probabilistically realisable* in a class $\mathcal{R}$ (PR($\mathcal{R}$)) if some member of $\mathcal{R}$ (all measures / stationary / supported on a subshift / memory-m Markov, an observer's commitment lattice) induces

*Research Computing Group, Simon Fraser University, Burnaby, BC, Canada. `pmahon@sfu.ca`

it. A *model* is *contextual* when it is EA but not $\mathrm{PR}(\mathcal{R})$. A protocol is *commensurable* when $\mathrm{EA} \Rightarrow \mathrm{PR}(\mathcal{R})$ for all coherent data.

The two protocol axes are *reporting* (window layout) and *coarsening* (compression); the commitment class $\mathcal{R}$ is the third, graded, axis.

The geometric reformulation. For a protocol on a finite outcome set, coherent data forms a polytope C (the local/marginal polytope in the sense of Wainwright–Jordan [17]) and realisable data a polytope $R = \mathrm{conv}\{v(x)\}$, the convex hull of the cell-incidence vectors of the global configurations. Always $R \subseteq C$.

Proposition 1.2 (commensurability is $C = R$)**.** *A protocol is commensurable iff $C = R$. A contextual model is exactly a point of $C \setminus R$, and (over rational data) any separating facet is a rational certificate of non-realisability.*

This makes the question exactly computable: C and R are enumerable in exact arithmetic, so every commensurability verdict is either a proof that $C = R$ or an exhibited witness in $C \setminus R$. We adopt throughout the discipline this enforces: a "commensurable" verdict is admitted only from a theorem or an exact enumeration, a "contextual" verdict only from an exhibited rational witness (§3.1).

Results. Section 2 proves the structure theory: the trichotomy locating all hardness in the constraint variety (§2.1), the parity theorem (§2.2), the taming catalogue with Circuit Localization for the symmetric loopless marginal-determined languages (§2.3, with the loopy and general-cyclic extensions verified to finite scale and marked as such), and the winding characterisation with its Lean-certified discriminant (§2.4). Section 3 states the open universal impossibility conjecture — its reduction to a combinatorial pruning lemma and the gap that reduction carries — and draws the reconstruction consequences.

What is new, and what is borrowed. The polyhedral facts this paper uses are classical and cited as such: the coherence/realisability polytopes are the local and marginal polytopes of graphical models [17], in the presheaf language of Abramsky–Brandenburger [2]; acyclic extendability is Vorob'ev's theorem [16]; the universality of the constrained case is De Loera–Onn [7] and, for the marginal polytope itself, Průša–Werner [14]; the parity mechanism is stable-set integrality (Nemhauser–Trotter [12], Chvátal [6], Hoffman–Kruskal [10], König); the taming mechanisms specialise total-unimodularity and perfect-graph theory. Our contribution is the *observational and dynamical* layer over this mathematics: the EA/PR framing of symbolic-dynamical protocols, the reading of a cyclic protocol as recurrence (so that contextuality of scope data occurs exactly on odd-period orbits), the winding / primitive-orbit criterion with its covering-space proof, and the assembly of the tamings into a localisation structure theory. This is a bridge over known polyhedral combinatorics; where it rises above translation is the winding criterion and the localisation assembly, which we have not found in the marginal-problem, contextuality, or constraint-satisfaction literatures — in particular the localisation line sits strictly above the satisfiability-width line (Barto–Kozik [4]) and must be read against the tightness criterion of Thapper–Živný [15], from which the recurrence/covering content distinguishes it.

2 The structure theory

2.1 The trichotomy

We first place the hardness of the general problem. Write a symbolic language as a relation ρ (the legal adjacent pairs of a shift of finite type); a protocol reports windows on the configurations legal for ρ.

Theorem 2.1 (acyclic reporting is tame; Vorob'ev). *If the reporting hypergraph of a protocol is acyclic, the protocol is commensurable for every language ρ.*

This is Vorob'ev's theorem in the present language: Vorob'ev [16] proves that a family of consistent marginals extends iff the underlying complex is *regular* (equivalently, acyclic), and acyclicity is the running-intersection / GYO condition of database theory [5]; the identification of the two is due to Barbosa (in Abramsky [1]). We include the statement for completeness and claim no novelty for it: the conditional-product glue over a junction tree assembles the window-marginals into a global distribution supported in the variety of ρ, so realisation costs no hypothesis on ρ. Frustration requires a cycle — which is where the rest of the paper lives.

Theorem 2.2 (full-support coordinate protocols are tame). *A k-context protocol of coordinate contexts with full product support is commensurable: the product measure realises any coherent margins.*

The wild region is the third. A protocol may carry a *constraint variety* — the configurations restricted by ρ — and there the local polytope inherits arbitrary structure.

Theorem 2.3 (variety-typed protocols are universal). *For protocols with three or more coordinate contexts subject to a constraint variety, the achievable coherence polytopes realise every rational polytope up to affine equivalence. Consequently: no finite facet taxonomy exists, model membership is* NP*-complete already at three contexts, and no bounded certificate normal form can classify contextual models in general.*

Proof sketch. By the covariance map [3] the relevant polytopes are cut/correlation polytopes, and by De Loera–Onn universality [7] every rational polytope is a face of a three-way transportation polytope, faces realised as imposed zeros ("punctures"), i.e. as constraint varieties. The NP-completeness floor is Pitowsky [13]. □

Theorem 2.3 is the reason the classification is posed at the *protocol* level (a $\forall\exists$ question over structured families) and not the model level: at the model level it is provably intractable. The remainder of the paper classifies protocol families where structure defeats the universality.

2.2 The parity theorem

The cleanest cyclic case is the golden-mean shift: ρ_{gm} forbids adjacent 1s. Observe it around a ring of length L (scope contexts = the L edge-windows of the cycle C_L).

Theorem 2.4 (parity). *The golden-mean ring protocol on C_L is commensurable iff L is even.*

Proof. Reduction. On the variety, each edge distribution is affine in the endpoint vertex-marginals ($x_i = 1$ forces $x_{i+1} = 0$), and adjacent coherence is exactly vertex-marginal matching. In marginal coordinates $C \cong \mathrm{FSTAB}(C_L)$ (the edge-constrained fractional stable-set polytope $\{u \geq 0,\ u_i + u_{i+1} \leq 1\}$) and $R \cong \mathrm{STAB}(C_L)$ (convex hull of independent-set indicators): homomorphisms $C_L \to$ golden-mean are exactly independent sets.

Parity. The incidence matrix of C_L has determinant $1 - (-1)^L$, and every proper square submatrix is forest-structured, hence totally unimodular. For L even the matrix is TU, so $\mathrm{FSTAB}(C_L)$ is integral and equals $\mathrm{STAB}(C_L)$: $C = R$, commensurable. For L odd a fractional vertex must have all coordinates positive (a zero coordinate lands on a path face, forest, TU, integral), forcing every edge constraint tight and, by nonsingularity, the unique fractional vertex $u \equiv \frac{1}{2}$. Thus $\mathrm{FSTAB}(C_L^{\mathrm{odd}}) = \mathrm{STAB}(C_L)$ plus exactly one extra vertex: the EA/PR gap on an odd golden-mean ring is a single point. □

Corollary 2.5 (the odd-ring PR-box). *The half-model $u \equiv \frac{1}{2}$ is separated by the odd-cycle inequality $\sum_i u_i \leq (L-1)/2$, with relative violation $1/(L-1)$. It is strongly contextual (contextual fraction 1): any noncontextual part would need $v_i + v_{i+1} \geq 1$ on every edge, giving $2\sum v_i \geq L > L - 1$. It is the golden-mean ring's analogue of a PR-box.*

Remark 2.6 (dynamical reading)**.** A cyclic scope structure is recurrence: sliding edge-windows on a period-L orbit identify time modulo L. Theorem 2.4 says golden-mean dynamics produces contextual scope data exactly on odd-period recurrence — alternation cannot close an odd ring, observed. The same odd-cycle facet is the KCBS/pentagon coverage bound [11], reached here by recurrence rather than by exclusivity: distinct observational routes to one polyhedral obstruction.

The parity spectrum varies with the language: the order relation $\{\leq\}$ is tame on all rings; XOR is tame on even rings (odd rings have empty variety); the full (unconstrained) language is tame on none. More constraint can mean more tame — the variety's rigidity can rescue commensurability the free world lacks.

2.3 Tamings and Circuit Localization

Each structured language that resists contextuality does so by a nameable mechanism. We isolate ten; the running discipline was to catalogue the *reason* a candidate wild protocol turned out tame, and to register each mechanism against a fresh prediction before testing it.

Definition 2.7 (the taming mechanisms)**.** A protocol/language is tamed by one of:

(1) *chains / empty-intersection* (§2.2 base case);

(2) *cross-context cliques*;

(3) *determinism*: a functional transfer relation;

(4) *grading*: the transfer digraph is fibered over $\mathbb{Z}_d$;

(5) *transient branching*: all branching lies off the recurrent core;

(6) *direct-sum decomposition*: a disconnected relation splits componentwise;

(7) *signability*: a column signing driven by the frame's bipartition giving total unimodularity, by the sufficient condition of Hoffman and Gale (appendix to [9]; cf. Ghouila-Houri [8]); König's theorem is the edge case;

(8) *twin expansion*: expanding a non-reflexive state into label-twins, realised by a constructive θ-sweep coupling;

(9) *monotone collapse*: a non-symmetric relation induces monotone potentials that strong connectivity flattens (the SCC-reduction lemma);

(10) *bipartite-target phase decoupling*: a bipartite-target language is a fibered join of two copies of its edge polytope over a global phase.

Tamings (7), (8), (9), (10) are proved theorems with machine-corroborated joints; (3)–(6) are structural reductions. The signable case is representative:

Theorem 2.8 (signable languages are tame on bipartite frames)**.** *Let a language be* marginal-determined *(its cell simplex is affine in the endpoint marginals) and* signable *(the edge-constraint matrix admits a $\{\pm 1\}$ column signing along the frame's bipartition). Then on every bipartite frame it is commensurable. The realisation is constructive: an anti-comonotone θ-sweep (threshold θ on one bipartition class, $1-\theta$ on the other) produces an exact mixture of global configurations reproducing any coherent point.*

Proof sketch. In marginal coordinates the edge-nonnegativity rows admit a Hoffman–Gale column signing exactly when the frame is 2-colourable, making the matrix totally unimodular (Hoffman–Gale appendix to [9]); integral right-hand sides then give integral vertices, so $C = R$. The θ-sweep is the explicit decomposition, a one-parameter alignment of the two bipartition classes. □

A counting fact then closes the loopless symmetric case entirely.

Theorem 2.9 (forest counting)**.** *A symmetric loopless language is marginal-determined iff, as a homomorphism target, it is a forest. Every such language phase-decouples on bipartite frames.*

Proof sketch. Marginal-determination requires $|\rho| \leq 2|A| - 1$; a symmetric loopless relation has $|\rho| = 2|E(H)|$, forcing $|E(H)| \leq |V(H)| - 1$, i.e. H a forest, hence bipartite; Theorem 2.8 and taming (10) apply. □

Theorem 2.10 (Circuit Localization, symmetric loopless case)**.** *Every symmetric loopless marginal-determined language is commensurable on every gate-passing frame. Contextuality is excluded by structure: such a language is a forest target (Theorem 2.9), tamed by phase-decoupling on bipartite frames and by acyclicity (Theorem 2.1) otherwise.*

Proof. By Theorem 2.9 a symmetric loopless marginal-determined language is, as a homomorphism target, a forest, hence bipartite. On a bipartite frame it is commensurable by phase-decoupling (taming (10), Definition 2.7), which reduces $C = R$ to the signable case (Theorem 2.8); on any frame, acyclic reporting is commensurable for every language (Theorem 2.1). The two cover the gate-passing frames. □

Remark 2.11 (the loopy and general cases: proved boundary)**.** The loopless restriction is where the theorem is *proved*. Two extensions were verified only to finite scale and are stated as such, not as theorems. First, the *loopy* symmetric marginal-determined languages: an exhaustive search finds no pair-bipartite non-signable survivor at ≤ 4 states (so the loopy remainder is tamed by signability, twin expansion, or phase-decoupling there), but no general argument yet rules out a survivor at larger alphabets. Second, the exclusion of *cyclic* symmetric targets rests on witness-certified frustration for the cycles $C_3, \ldots, C_6$ to bounded layered depth; the all-target, all-length statement is established only for golden-mean odd rings (Theorem 2.4). The full symmetric class is therefore commensurable through the verified scale, with the loopy and general-cyclic steps owed; the honest headline is the loopless theorem above.

2.4 The winding characterisation and its certified discriminant

Underlying the cyclic analysis is a structural characterisation. Lift a length-L ring protocol to its *layered graph*: vertices (i, a) for layer $i \in \mathbb{Z}_L$ and state a, arcs $(i, a) \to (i{+}1, b)$ for legal pairs. A $\{0, 1\}$-valuation is contextual iff the coherence polytope has a fractional vertex, and by flow decomposition the fractional vertices are the normalised simple directed cycles of winding ≥ 2.

Theorem 2.12 (winding characterisation)**.** *Under the genericity gate (non-adjacent contexts share only trivial events), a length L is unsafe for ρ iff the layered ring over $\mathbb{Z}_L$ carries a layer-injective simple cycle of winding ≥ 2. The characterisation is structural; only its eventual-periodicity refinement (§3) is conjectural.*

The proof turns on a discriminant that informal reasoning repeatedly mishandled: a closed walk that "closes up" need not be a *simple* cycle, and the exact certificate of simplicity is layer-injectivity (each layer visited in distinct states).

Lemma 2.13 (layer-injectivity discriminant; Lean-certified)**.** *A layered-ring closed walk is a simple cycle (its vertex map is injective) iff it is layer-injective (its state map is injective on each layer fibre).*

This lemma is machine-verified in Lean 4 (`formalization/QuerySystem/QuerySystem/WindingInjectivi` statements `simple_iff_layerInjective`, `simple_of_layerInjective`, `not_simple_of_collision`), with zero `sorry`s, axioms exactly `propext` and `Quot.sound`, and non-vacuity examples (a winding-2 collision proved non-simple, a layer-injective trace proved simple). It is the first machine-certified result of the reconstruction structure theory, and deliberately independent of the open material of §3: pure vertex-trace combinatorics.

3 The open case and consequences

The symmetric world is settled (§2.3). The residence of any remaining contextual protocol is the strongly-connected non-symmetric class. An SCC-reduction lemma (taming (9), monotone collapse) reduces commensurability wholesale to strongly connected languages; exhaustive search finds no contextual protocol there at small scale (none at ≤ 4 states; none through 9 edges at 5 states, at certificate grade). We record the conjecture and its status honestly.

Conjecture 3.1 (universal impossibility)**.** The fork does not exist: Circuit Localization holds universally for total languages on gate-passing, time-realisable frames. Equivalently, blocking the frustrating exchange at a rich set of lengths forces one of the catalogued tamings, each of which localises the polytope.

The intended proof runs through eventual periodicity of the unsafe set (a finite union of arithmetic progressions, Wielandt-bounded), which reduces to a combinatorial *pruning lemma*.

Conjecture 3.2 (pruning lemma)**.** Let TR_k be the lengths admitting a diagonal-avoiding tuple-rotation walk in the k-fold tensor power, and LISC_k the lengths admitting a layer-injective simple cycle of winding k. Then $\mathrm{TR}_k = \mathrm{LISC}_k$ cofinitely.

The inclusion $\mathrm{LISC}_k \subseteq \mathrm{TR}_k$ is immediate. The reverse reduces, at $k = 2$, to a flow-decomposition statement (a winding-2 circulation is a single winding-2 cycle iff its two strands are exchange-linked); we record explicitly that this reduction currently drops a *relative-phase* degree of freedom between the strands (a winding-2 simple cycle interleaves its wraps at a phase a lockstep exchange cannot express), so the corrected $k = 2$ residual is phase-parametrised strand-disjointness. Empirically no $\mathrm{TR}_k \setminus \mathrm{LISC}_k$ disagreement appears across an adversarial stress set to layered length 35 (including validated-imprimitive digraphs, where the natural counterexample was predicted and did not occur), which is evidence, not proof. The conjecture is open; its proof is not to be assembled until the pruning lemma is settled structurally.

3.1 Consequences for reconstruction

The classification licenses a reconstruction guarantee, scoped honestly.

The tame classes give a decidable guarantee. For acyclic reporting (Theorem 2.1), for symmetric marginal-determined languages (Theorem 2.10), and for even-cyclic golden-mean-type recurrence (Theorem 2.4), coherent window-statistics *do* determine a global process, and the winding characterisation (Theorem 2.12) supplies a decidable criterion — the presence of a layer-injective winding-≥ 2 cycle — separating the safe protocols from the contextual ones. This is the licensing layer beneath process reconstruction ("statistical Takens": recovery of a measure-plus-shift, not a manifold), with an explicit boundary on when reconstruction is possible.

The boundary is intrinsic, not accidental. By Theorem 2.3, general variety-typed protocols are as hard as arbitrary rational polytope membership; contextuality in observation is not a pathology to be engineered away but an intrinsic feature of observing-through- windows, present exactly when the observation geometry admits the winding obstruction. Where Conjecture 3.1 holds, the guarantee is unconditional for total languages: no protocol over such a world manufactures contextual data unless the decidable exchange criterion fires.

One phenomenon, three faces. The failure of local coherence to globalise appears here dynamically, in Paper II algebraically (dispersion-free states and the collapse of PR on non-distributive logics), and in the σ-essential witness set-theoretically (a finitely-coherent pattern with no σ-additive global state). The dynamical statement is the sharpest: *frustration is an unwitnessed exchange — two histories that trade places around the clock of observation without meeting — and contextual data is the shadow of an unresolved covering of time.*

Note on method and verification

Every commensurability verdict in §§2.1–**??** is derived from a theorem or an exact rational enumeration; every contextuality verdict from an exhibited rational witness in $C \setminus R$; sampling-based checks are used only as screening and never as the basis of a "commensurable" claim. Lemma 2.13 is machine-checked in Lean 4. The constructive tamings (θ-sweep, phase-mixture, monotone collapse, empty-intersection) are the queued targets for further certification. The open material of §3 is stated as conjecture with its reduction gap named.

References

[1] Samson Abramsky. Relational databases and Bell's theorem. In *In Search of Elegance in the Theory and Practice of Computation*, volume 8000 of *Lecture Notes in Computer Science*, pages 13–35. Springer, 2013.

[2] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[3] David Avis, Hiroshi Imai, Tsuyoshi Ito, and Yuuya Sasaki. Two-party Bell inequalities derived from combinatorics via triangular elimination. *Journal of Physics A: Mathematical and General*, 38(50):10971–10987, 2005.

[4] Libor Barto and Marcin Kozik. Constraint satisfaction problems solvable by local consistency methods. *Journal of the ACM*, 61(1):3:1–3:19, 2014.

[5] Catriel Beeri, Ronald Fagin, David Maier, and Mihalis Yannakakis. On the desirability of acyclic database schemes. *Journal of the ACM*, 30(3):479–513, 1983.

[6] Vašek Chvátal. On certain polytopes associated with graphs. *Journal of Combinatorial Theory, Series B*, 18(2):138–154, 1975.

[7] Jesús A. De Loera and Shmuel Onn. The complexity of three-way statistical tables. *SIAM Journal on Computing*, 33(4):819–836, 2004.

[8] Alain Ghouila-Houri. Caractérisation des matrices totalement unimodulaires. *Comptes Rendus de l'Académie des Sciences*, 254:1192–1194, 1962.

[9] Isidore Heller and C. B. Tompkins. An extension of a theorem of Dantzig's. *Annals of Mathematics Studies*, 38:247–254, 1956. Total-unimodularity sufficient condition in the appendix by A. J. Hoffman and D. Gale.

[10] Alan J. Hoffman and Joseph B. Kruskal. Integral boundary points of convex polyhedra. *Annals of Mathematics Studies*, 38:223–246, 1956.

[11] Alexander A. Klyachko, M. Ali Can, Sinem Binicioğlu, and Alexander S. Shumovsky. Simple test for hidden variables in spin-1 systems. *Physical Review Letters*, 101(2):020403, 2008.

[12] George L. Nemhauser and Leslie E. Trotter. Properties of vertex packing and independence system polyhedra. *Mathematical Programming*, 6(1):48–61, 1974.

[13] Itamar Pitowsky. Correlation polytopes: their geometry and complexity. *Mathematical Programming*, 50(1–3):395–414, 1991.

[14] Daniel Průša and Tomáš Werner. Universality of the local marginal polytope. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 37(4):898–904, 2015.

[15] Johan Thapper and Stanislav Živný. The complexity of finite-valued CSPs. *Journal of the ACM*, 63(4):37:1–37:33, 2016.

[16] N. N. Vorob'ev. Consistent families of measures and their extensions. *Theory of Probability and Its Applications*, 7(2):147–163, 1962.

[17] Martin J. Wainwright and Michael I. Jordan. *Graphical Models, Exponential Families, and Variational Inference*, volume 1 of *Foundations and Trends in Machine Learning*. Now Publishers, 2008.